

EX-NO:06**MULTIMEDIA DATABASE****03-11-23****AIM:**

To Create a Database to store Multimedia elements and perform data retrieval operations.

CREATE COMMAND:

```
SQL> CREATE TABLE Multimedia (Id NUMBER GENERATED ALWAYS AS IDENTITY  
PRIMARY KEY, Title VARCHAR2(255), File path VARCHAR2(255),  
Type VARCHAR2(50) CHECK (type IN ('image', 'video', 'audio')), Description CLOB,  
Upload_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP);
```

```
Table created.
```

INSERT:

```
SQL> INSERT INTO Multimedia (title, file_path, type, description) VALUES ('Nature  
Image', 'C:\Users\Janani\Pictures\Saved Pictures\nature1.jpg', 'image', 'Beautiful Tree');
```

```
1 row created.
```

```
SQL> INSERT INTO Multimedia (title, file_path, type, description) values
```

```
('Butterfly Image', 'C:\Users\Janani\Pictures\Saved Pictures\nature2.jpg', 'image', 'Electric  
Butterfly');
```

```
1 row created.
```

```
SQL> INSERT INTO Multimedia (title, file_path, type, description) values
```

```
('Dolphin', 'C:\Users\Janani\Pictures\Saved Pictures\Dolphin.jpg', 'image', 'Flying Dolphin');
```

```
1 row created.
```

```
SQL> INSERT INTO Multimedia (title, file_path, type, description) values
```

```
('FS-PPT', 'C:\Users\Janani\Desktop\Janani\PG Books\Full Stack Exercises\React  
Components\Fullstack-Presentation.mp4', 'video', 'PPT');
```

```
1 row created.
```

```
SQL> INSERT INTO Multimedia (title, file_path, type, description) values
('Youjennie', 'C:\Users\Janani\Desktop\Janani\songs\JENNIE - You jenniemp3', 'audio',
'You ');
```

```
1 row created.
```

AFTER INSERTING DISPLAY THE VALUES:

```
SQL> select * from multimedia;
```

ID	TITLE	FILE_PATH	TYPE	DESCRIPTION	UPLOAD_DATE
1	Nature Image	C:\Users\Janani\Pictures\Saved Pictures\nature1.jpg	image	Beautiful Tree	24-NOV-23 05.56.59.149000 PM
3	Butterfly Image	C:\Users\Janani\Pictures\Saved Pictures\nature2.jpg	image	Electric Butterfly	24-NOV-23 06.05.29.236000 PM
4	Dolphin	C:\Users\Janani\Pictures\Saved Pictures\Dolphin.jpg	image	Flying Dolphin	24-NOV-23 06.06.52.443000 PM
5	FS-PPT	C:\Users\Janani\Desktop\Janani\PG Books\Full Stack video Exercises\React Components\Fullstack-Presentation.mp4	video	PPT	24-NOV-23 06.08.01.953000 PM
6	Youjennie	C:\Users\Janani\Desktop\Janani\songs\JENNIE - You jenniemp3	audio	You	24-NOV-23 06.10.53.175000 PM

UPDATE:

```
SQL> UPDATE Multimedia SET title = 'Jennie',
file_path = 'C:\Users\Janani\Desktop\Janani\songs\JENNIE - You and Me.mp3',
description = 'You & Me' WHERE id = 6;
```

```
1 row updated.
```

AFTER UPDATE DISPLAY THE VALUES:

```
SQL> select * from multimedia;
```

ID	TITLE	FILE_PATH	TYPE	DESCRIPTION	UPLOAD_DATE
1	Nature Image	C:\Users\Janani\Pictures\Saved Pictures\nature1.jpg	image	Beautiful Tree	24-NOV-23 05.56.59.149000 PM
3	Butterfly Image	C:\Users\Janani\Pictures\Saved Pictures\nature2.jpg	image	Electric Butterfly	24-NOV-23 06.05.29.236000 PM
4	Dolphin	C:\Users\Janani\Pictures\Saved Pictures\Dolphin.jpg	image	Flying Dolphin	24-NOV-23 06.06.52.443000 PM
5	FS-PPT	C:\Users\Janani\Desktop\Janani\PG Books\Full Stack video Exercises\React Components\Fullstack-Presentation.mp4	video	PPT	24-NOV-23 06.08.01.953000 PM
6	Jennie	C:\Users\Janani\Desktop\Janani\songs\JENNIE - You and Me.mp3	audio	You & Me	24-NOV-23 06.10.53.175000 PM

RETRIEVE MULTIMEDIA ELEMENTS OF A SPECIFIC TYPE:

SQL> SELECT * FROM Multimedia WHERE type = 'image';

ID	TITLE	FILE_PATH	TYPE	DESCRIPTION	UPLOAD_DATE
1	Nature Image	C:\Users\Janani\Pictures\Saved Pictures\nature1.jpg	image	Beautiful Tree	24-NOV-23 05.56.59.149000 PM
3	Butterfly Image	C:\Users\Janani\Pictures\Saved Pictures\nature2.jpg	image	Electric Butterfly	24-NOV-23 06.05.29.236000 PM
4	Dolphin	C:\Users\Janani\Pictures\Saved Pictures\Dolphin.jpg	image	Flying Dolphin	24-NOV-23 06.06.52.443000 PM

SQL> SELECT * FROM Multimedia WHERE type = 'video';

ID	TITLE	FILE_PATH	TYPE	DESCRIPTION	UPLOAD_DATE
5	FS-PPT	C:\Users\Janani\Desktop\Janani\PG Books\Full Stack video Exercises\React Components\Fullstack-Presentation.mp4	video	PPT	24-NOV-23 06.08.01.953000 PM

SQL> SELECT * FROM Multimedia WHERE type = 'audio';

ID	TITLE	FILE_PATH	TYPE	DESCRIPTION	UPLOAD_DATE
6	Jennie	C:\Users\Janani\Desktop\Janani\songs\JENNIE - You and Me.mp3	audio	You & Me	24-NOV-23 06.10.53.175000 PM

SQL> SELECT * FROM Multimedia WHERE DBMS_LOB.INSTR(Description, 'PPT') > 0;

ID	TITLE	FILE_PATH	TYPE	DESCRIPTION	UPLOAD_DATE
5	FS-PPT	C:\Users\Janani\Desktop\Janani\PG Books\Full Stack video Exercises\React Components\Fullstack-Presentation.mp4	video	PPT	24-NOV-23 06.08.01.953000 PM

RETRIEVE MULTIMEDIA ELEMENTS WITH A SPECIFIC TITLE:

SQL> SELECT * FROM Multimedia WHERE UPPER(title) = 'NATURE IMAGE';

ID	TITLE	FILE_PATH	TYPE	DESCRIPTION	UPLOAD_DATE
1	Nature Image	C:\Users\Janani\Pictures\Saved Pictures\nature1.jpg	image	Beautiful Tree	24-NOV-23 05.56.59.149000 PM

MULTIMEDIA ELEMENTS UPLOADED ON A SPECIFIC DATE:

SQL> SELECT * FROM Multimedia WHERE TO_CHAR(upload_date, 'DD-MON-YY') = '24-NOV-23';

ID	TITLE	FILE_PATH	TYPE	DESCRIPTION	UPLOAD_DATE
1	Nature Image	C:\Users\Janani\Pictures\Saved Pictures\nature1.jpg	image	Beautiful Tree	24-NOV-23 05.56.59.149000 PM
3	Butterfly Image	C:\Users\Janani\Pictures\Saved Pictures\nature2.jpg	image	Electric Butterfly	24-NOV-23 06.05.29.236000 PM
4	Dolphin	C:\Users\Janani\Pictures\Saved Pictures\Dolphin.jpg	image	Flying Dolphin	24-NOV-23 06.06.52.443000 PM
5	FS-PPT	C:\Users\Janani\Desktop\Janani\PG Books\Full Stack video Exercises\React Components\Fullstack-Presentation.mp4	video	PPT	24-NOV-23 06.08.01.953000 PM
6	Jennie	C:\Users\Janani\Desktop\Janani\songs\JENNIE - You and Me.mp3	audio	You & Me	24-NOV-23 06.10.53.175000 PM

MULTIMEDIA ELEMENTS UPLOADED AFTER A SPECIFIC TIME:

SQL> SELECT * FROM Multimedia WHERE TO_DATE (upload_date, 'DD-MON-YY HH24:MI: SS') > TO_DATE ('24-NOV-23 06:00:00', 'DD-MON-YY HH24:MI: SS');

```
no rows selected
```

RETRIEVE MULTIMEDIA ELEMENTS WITH DESCRIPTIONS LONGER THAN A CERTAIN LENGTH:

SQL> SELECT * FROM Multimedia WHERE LENGTH(description) > 10;

ID	TITLE	FILE_PATH	TYPE	DESCRIPTION	UPLOAD_DATE
1	Nature Image	C:\Users\Janani\Pictures\Saved Pictures\nature1.jpg	image	Beautiful Tree	24-NOV-23 05.56.59.149000 PM
3	Butterfly Image	C:\Users\Janani\Pictures\Saved Pictures\nature2.jpg	image	Electric Butterfly	24-NOV-23 06.05.29.236000 PM
4	Dolphin	C:\Users\Janani\Pictures\Saved Pictures\Dolphin.jpg	image	Flying Dolphin	24-NOV-23 06.06.52.443000 PM

AGGREGATE OPERATIONS:**COUNT:**

SQL> SELECT COUNT (*) FROM Multimedia;

```
COUNT(*)
-----
5
```

SQL> SELECT COUNT (*) FROM Multimedia WHERE type = 'image';

```
COUNT(*)
-----
3
```

SQL> SELECT type, COUNT (*) AS count FROM Multimedia GROUP BY type;

```
TYPE          COUNT
-----
image         3
video         1
audio         1
```

SUM:

SQL> SELECT SUM(ID) FROM Multimedia;

```
SUM(ID)
-----
19
```

SQL> SELECT SUM(LENGTH(description)) FROM Multimedia;

```
SUM(LENGTH(DESCRIPTION))
-----
57
```

SQL> SELECT type, SUM(1) AS total_items FROM Multimedia GROUP BY type;

```
TYPE          TOTAL_ITEMS
-----
image          3
video          1
audio          1
```

AVERAGE:

SQL> SELECT AVG(ID) FROM Multimedia;

```
AVG(ID)
-----
3.8
```

SQL> SELECT AVG(EXTRACT(MINUTE FROM upload_date)) AS
average_upload_minute FROM Multimedia;

```
AVERAGE_UPLOAD_MINUTE
-----
17
```

MAXIMUM AND MINIMUM:

SQL> SELECT MAX(ID), MIN(ID) FROM Multimedia;

```
MAX(ID)      MIN(ID)
-----
6            1
```

SQL> SELECT MAX(upload_date) AS max_upload_date, MIN(upload_date) AS
min_upload_date FROM Multimedia;

```
MAX_UPLOAD_DATE          MIN_UPLOAD_DATE
-----
24-NOV-23 06.10.53.175000 PM 24-NOV-23 05.56.59.149000 PM
```

RETRIEVE MULTIMEDIA ELEMENTS WITH MAXIMUM UPLOAD DATE:

SQL> SELECT * FROM Multimedia WHERE

upload_date = (SELECT MAX(upload_date) FROM Multimedia);

ID	TITLE	FILE_PATH	TYPE	DESCRIPTION	UPLOAD_DATE
6	Jennie	C:\Users\Janani\Desktop\Janani\songs\JENNIE - You and Me.mp3	audio	You & Me	24-NOV-23 06.10.53.175000 PM

CALCULATE THE AVERAGE LENGTH OF DESCRIPTIONS FOR EACH TYPE:

SQL> SELECT type, AVG(LENGTH(description)) AS avg_description_length FROM Multimedia GROUP BY type;

TYPE	AVG_DESCRIPTION_LENGTH
image	15.3333333
video	3
audio	8

RESULT:

Thus, creation of Database to store Multimedia elements and perform data retrieval operations are done successfully.

OBSERVATION	RECORD	SIGNATURE